

A Semantic Boundary for Concern-Weighted Metric Deformation in Pretrained Transformers

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Research-Derived Experiments · Paper B Externality Follow-up

Abstract

Paper B found that a movable spatial loss-weight field, operationally called concern, causally moves metric-density deformation in path-integration RNN, Transformer, and JEPA-style models. This follow-up tests the strongest immediate limitation: does the effect generalize to real text semantics in pretrained transformer encoders? We pre-registered a 20 Newsgroups moved-target experiment with two pretrained encoders (DistilBERT and all-MiniLM-L6-v2), two objectives (classifier and JEPA-like predictive latent training), four semantic targets, random-matched weighting controls, and a 2% standard-error gate on primary causal effects. Across 9,216 Modal H200/H100 fine-tuning cells and 4,096 paired effects, the confirmatory gate fails decisively: the pooled semantic-margin lift is -0.441 vs uniform [-0.455, -0.427], SE 0.007, and -0.441 vs random-matched controls [-0.455, -0.428], SE 0.007. Every family is negative with primary SE below 2%. Companion probes show real geometric movement--centroid separation, kNN purity, and often effective rank increase--but held-out target F1 and the registered local semantic margin decrease. The honest conclusion is not that Paper B is wrong; it is that the spatial metric-density claim does not yet transport to a pretrained text setting. This is a publishable boundary condition and a better foundation for the next paper than an overbroad generalization claim.

1. Why This Follow-up Exists

The original Paper B result is causal and clean inside its domain: move an external loss-weight field in a spatial path-integration task, and the learned metric-density peak moves with it. But the remaining reviewer objection is obvious and fair. The task is synthetic and spatial. Architecture variants named Transformer and JEPA do not by themselves prove that pretrained language or general AI systems allocate semantic resolution in the same way.

This experiment was designed to attack that limitation directly. The target is no longer a coordinate in a square arena. It is one of four real text classes: comp.graphics, rec.sport.hockey, sci.med, and sci.space from 20 Newsgroups. The intervention is still external and causal: upweight the training loss for one registered semantic class, move which class receives that weight, and test whether the representation-geometry change moves to that class.

Semantic follow-up: a real-text moved-target test, not a spatial coordinate bump

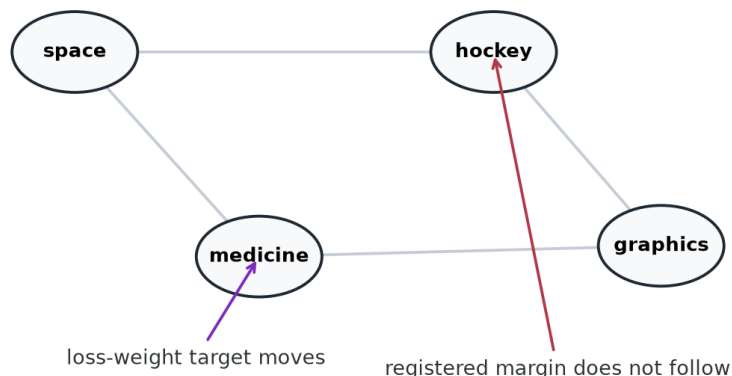


Figure 1. Schematic only. The semantic experiment moves a loss-weight target over text classes rather than over physical coordinates. The registered question is whether the local semantic margin follows that target.

2. Pre-registered Gate

Concern is used here only as operational shorthand for a scalar loss weight. The primary metric is local semantic margin in the learned latent: mean k-nearest different-class cosine distance minus mean k-nearest same-class cosine distance, z-scored across classes inside the same model. A family passes only if the concern-target margin lift is positive against both a uniform model and a random-matched weighting control, the specificity interval is positive, the primary standard errors are at most 0.02, the target margin rank is above chance, and the run uses real 20 Newsgroups data rather than a synthetic fallback.

The model families are DistilBERT and all-MiniLM-L6-v2, each with a classifier objective and a JEPA-like text objective. The JEPA-like variant predicts the stop-gradient latent of another same-class text example. It is a joint-embedding predictive analogue, not a claim to reproduce the official I-JEPA vision architecture.

3. Main Result: The Gate Fails

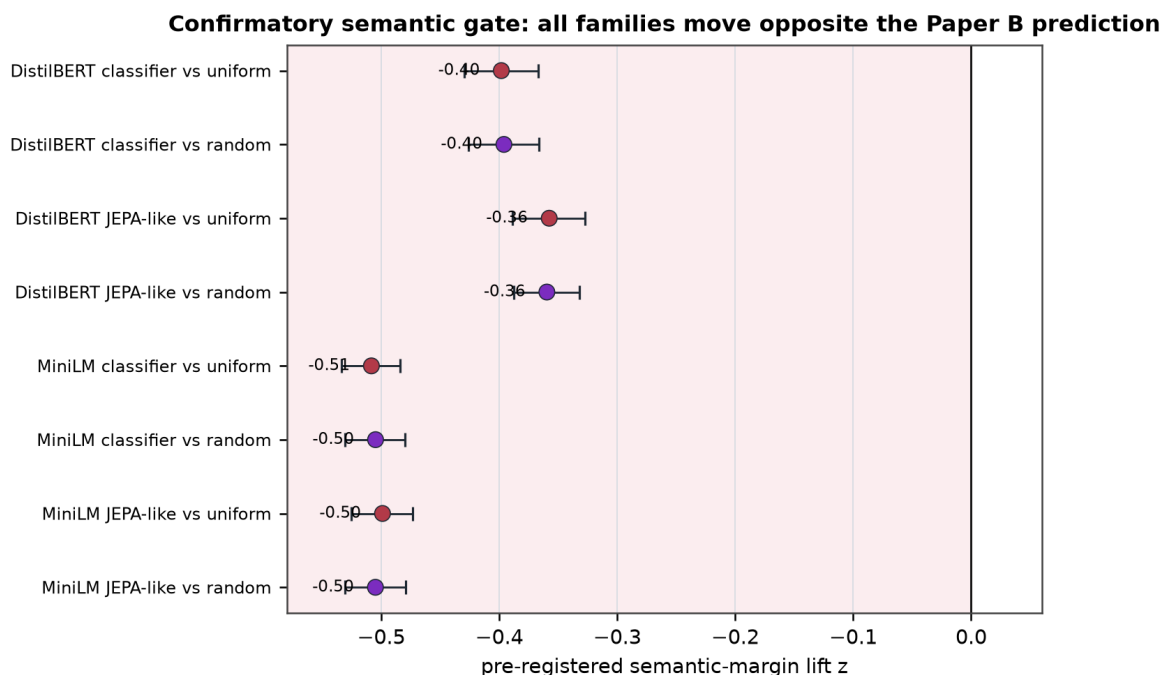


Figure 2. Primary confirmatory result. All four pretrained text families move opposite the registered Paper B semantic-margin prediction, against both uniform and random-matched controls. Error bars are bootstrap 95% intervals over seed-target effects.

Family	lift vs uniform	SE	lift vs random	SE	rank	gate
DistilBERT classifier	-0.398 [-0.430,-0.367]	0.016	-0.396 [-0.426,-0.366]	0.015	0.473	fail
DistilBERT JEPA-like	-0.358 [-0.389,-0.327]	0.015	-0.360 [-0.388,-0.332]	0.014	0.485	fail
MiniLM classifier	-0.508 [-0.534,-0.484]	0.013	-0.505 [-0.531,-0.480]	0.013	0.450	fail
MiniLM JEPA-like	-0.499 [-0.525,-0.474]	0.013	-0.505 [-0.531,-0.480]	0.013	0.452	fail

Table 1. Confirmatory gate. All primary standard errors are below the accepted 2% threshold, but all signs are negative.

The architecture-balanced pooled primary effect is -0.441 vs uniform with SE 0.007 and -0.441 vs random-matched controls with SE 0.007. The real-dataset gate passes; the causal semantic-margin gate does not. This is a precise negative result, not an ambiguous underpowered null.

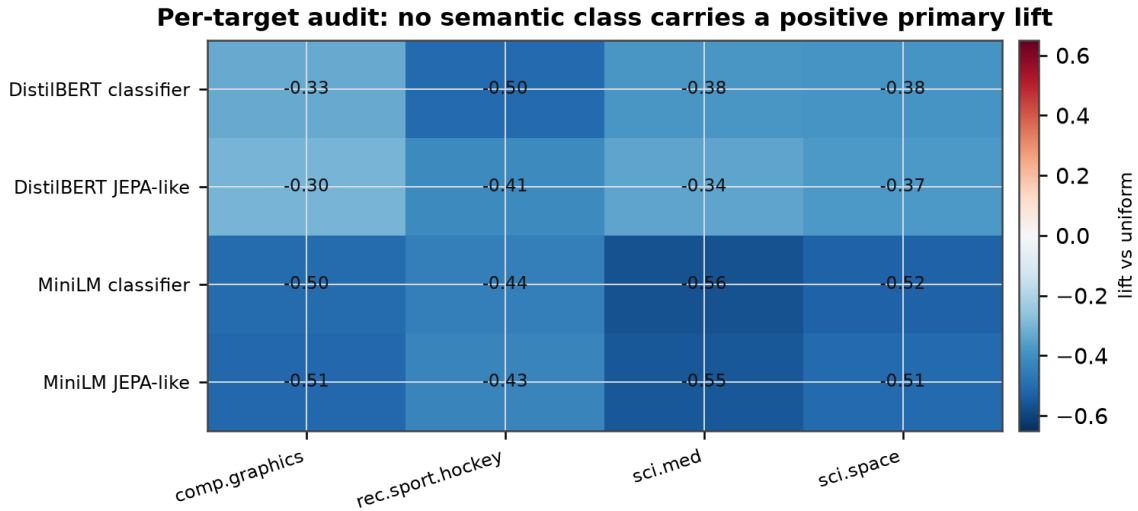


Figure 3. Per-target audit. No registered semantic class carries a positive primary lift; the result is not caused by one difficult topic.

4. Companion Geometry: A Split, Not a Vacuum

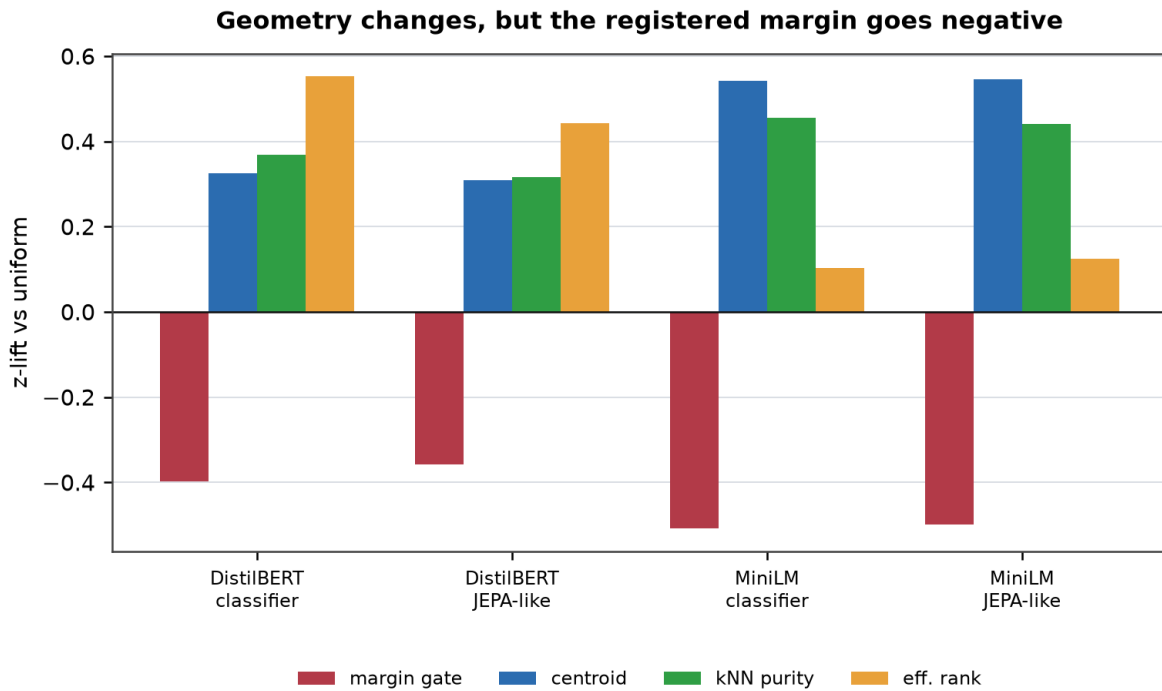


Figure 4. Companion metrics reveal a split geometry. Upweighting increases centroid separation and neighborhood purity, and often effective rank, even while the registered local semantic-margin gate becomes negative.

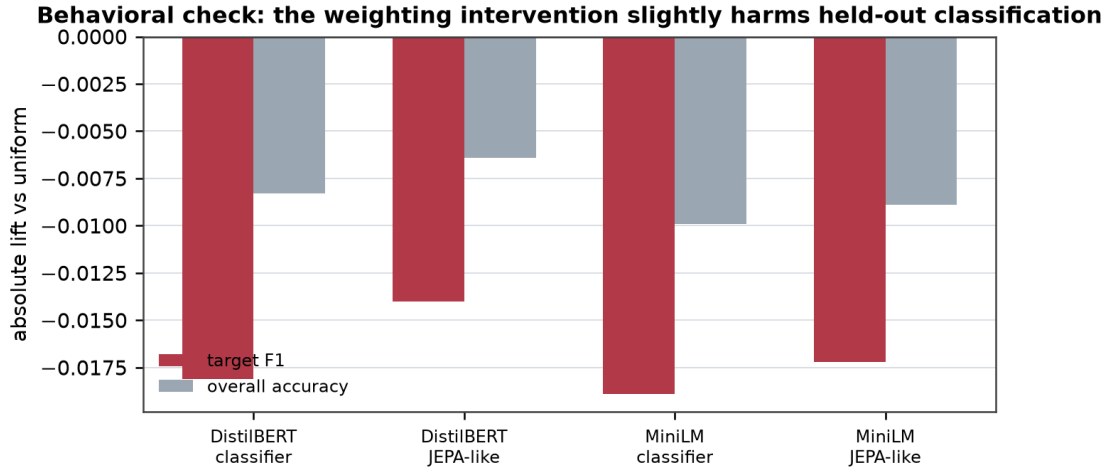


Figure 5. Behavioral check. The same intervention slightly reduces held-out target F1 and overall accuracy, so the companion geometry should not be rebranded as a successful task-performance effect.

This split matters. A careless analysis could have replaced the failed primary metric with the positive centroid or purity probes and declared victory. The pre-registration prevents that. What we can say is narrower and more useful: semantic loss weighting does reorganize the latent space, but in these pretrained text encoders it does not increase the registered local margin around the valued class. It appears closer to class-level re-centering or cluster sharpening, with a small behavioral cost, than to the spatial metric-density allocation observed in Paper B.

5. Consequences for Paper B

This result strengthens Paper B by limiting it. The spatial finding remains a robust causal mechanism result: within finite-capacity path integration, a moved value field moves the learned metric. But it should not be marketed as already established in pretrained language models or foundation-model settings. The correct NeurIPS/ACL-ready posture is: the spatial mechanism is real, the semantic transformer generalization is not yet confirmed, and the first large externality test found a boundary condition.

The next experiment should therefore not merely add more seeds. It should change the semantic task so that 'resolution' has a task-native meaning: retrieval under asymmetric cost, pairwise paraphrase discrimination inside the target class, or a contrastive corpus where valued examples require finer distinctions rather than broader class separation. Only then should a new confirmatory gate be frozen.

6. Limitations

The experiment uses four 20 Newsgroups topics, two compact pretrained encoders, and a JEPA-like text objective rather than a production-scale language model or official I-JEPA. The primary metric is a plausible semantic analogue of local metric density, but the positive companion results show that geometry has multiple non-equivalent notions. The result therefore rejects one transported claim, not all possible semantic concern hypotheses.

References

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